

Solutions to Fourth Problem Sheet

on Complex Analysis

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Topic: Complex Differentiability

"Complex analysis is the good twin and real analysis the evil one: beautiful formulas and elegant theorems seem to blossom spontaneously in the complex domain."
-Peter Henrici

"Real analysis is the battlefield; complex analysis is the paradise."
-Paul Halmos

"The miracle of complex differentiability is that from one simple condition — the Cauchy-Riemann equations — an entire universe of rigidity and beauty emerges."
-Lars Ahlfors

"In the complex domain, differentiability once implies differentiability infinitely many times. Nature is more generous than we deserve."
-Henri Cartan

"The requirement of complex differentiability is so rigid, so demanding, that functions satisfying it reveal the deepest secrets of mathematics."
-Carl Ludwig Siegel

"Riemann's concept of the complex derivative unified geometry, analysis and physics in one stroke, an act of mathematical genius that may never be surpassed."
-Felix Klein

Problem 1

Show that $f(z)$ is holomorphic if and only if $(f(z^*))^*$ is holomorphic.

Solution 1

($\Rightarrow$) If f is holomorphic, then by the Cauchy-Riemann equations we have

$$\partial_x u(x, y) = \partial_y v(x, y) \quad \text{and} \quad \partial_y u(x, y) = -\partial_x v(x, y).$$

On the other hand, $f^*(z^*) = u(x, -y) - iv(x, -y)$ so that

$$\partial_x(\Re(f)) = \partial_x u(x, y), \quad \partial_y(\Re(f)) = -\partial_y u(x, y), \quad \partial_x(\Im(f)) = -\partial_x v(x, y), \quad \partial_y(\Im(f)) = \partial_y v(x, y),$$

whence the conclusion is immediate. The converse also follows from the previous equation. $\star$

Problem 2

Let f and g be two functions such that $f(z_0) = g(z_0) = 0$, both differentiable at z_0 with $g'(z_0) \neq 0$. Prove that

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}.$$

The latter is known as the **L'Hospital's rule** for complex functions.

Solution 2

Suppose that $f(z_0) = g(z_0) = 0$, then

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{g(z) - g(z_0)} = \lim_{z \rightarrow z_0} \frac{\frac{f(z) - f(z_0)}{z - z_0}}{\frac{g(z) - g(z_0)}{z - z_0}},$$

and assuming differentiability of f, g at z_0 , the desired result now follows readily. $\star$

Problem 3

We say that a function f is **anitholomorphic** if it is the complex conjugate of a holomorphic function.

- i) Show that an anitholomorphic functions satisfies the **Anti-Cauchy-Riemann equations** (also known as the **conjugate Cauchy-Riemann equations**)

$$\partial_x u = -\partial_y v \quad \text{and} \quad \partial_y u = \partial_x v.$$

- ii) Is the converse also true?
iii) Can a complex function be holomorphic and anitholomorphic at the same time? If so, determine which type of functions, and if not, prove it.

Solution 3

- i) The proof is straightforward.
ii) Yes, if $f = u + iv$ satisfies the conjugate Cauchy-Riemann equations, then $g = u - iv \equiv \Re(g) + i\Im(g)$ is such that

$$\partial_x \Re(g) = \partial_y \Im(g) \quad \text{and} \quad \partial_y \Re(g) = -\partial_x \Im(g).$$

- iii) The two conditions combined can be shown to imply that $\partial_y v = \partial_y u = \partial_x u = \partial_x v = 0$ so that the only functions satisfying both are the constant functions. $\star$

Problem 4

Let $\mathcal{V} = C^2(U)$ with $U \subseteq \mathbb{R}$ be the linear space of functions $f : U \rightarrow \mathbb{R}$ at least class C^2 . We define the **Laplace operator** $\Delta : C^2(U) \rightarrow C(U)$ as

$$\Delta \doteq \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

We say that a function $f \in \mathcal{V}$ is **harmonic** if it is annihilated by the Laplace operator.

Show that the Cauchy-Riemann operators factorise the Laplace operator via

$$\Delta = 4 \cdot \partial_z \partial_{z^*},$$

and thus deduce that any holomorphic function is harmonic. Conclude that the sum of a holomorphic function with an anitholomorphic function yields a harmonic function. Finally, write $z = x + ict$ and hence show that the general solution to the one-dimensional wave equation is given by

$$u(x, t) = f(x + ct) + g(x - ct),$$

with f, g any two differentiable functions.

Solution 4

Compute

$$\partial_z \partial_{z^*} = \frac{1}{2} (\partial_x - i\partial_y) \frac{1}{2} (\partial_x + i\partial_y) = \frac{1}{4} (\partial_x^2 + \partial_y^2 + i\partial_x \partial_y - i\partial_y \partial_x),$$

it is a well-known fact (Clairaut's theorem) that $[\partial_x, \partial_y] = 0$ on $C^2(U)$ so that

$$\partial_z \partial_{z^*} = \frac{1}{4} \Delta \iff \Delta = 4 \cdot \partial_z \partial_{z^*}.$$

Recall that f is holomorphic on U if and only if $\partial_{z^*} f = 0$ on U so that, in particular, if f is holomorphic, then $\partial_z \partial_{z^*} f = \partial_z 0 = 0$ and thus the conclusion easily follows from the factorisation of the Laplacian. An antiholomorphic function is annihilated by ∂_z so that it is also harmonic due to the factorisation. If we write $z = x + ict$ then the sum of holomorphic plus antiholomorphic can be written as

$$f(x + ict) + g(x - ict),$$

which is a hint that $f(x + ct) + g(x - ct)$ is harmonic, in fact, we can compute the partial derivatives and it is straightforward to see that

$$\frac{\partial^2 f}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 f}{\partial c^2} = 0 \quad \text{and} \quad \frac{\partial^2 g}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 g}{\partial c^2} = 0.$$



Problem 5

By considering the gradients of the real and imaginary parts of a holomorphic function, show that the contour lines of the former and the latter intersect orthogonally. This result is represented graphically in Figure 1.

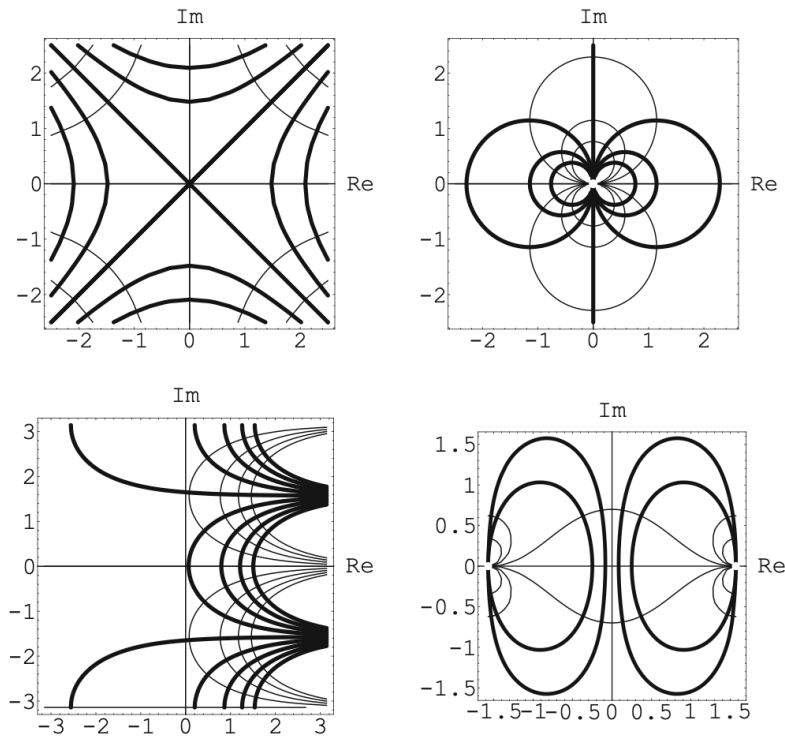


Figure 1: Orthogonal intersections of the real (thick trace) and imaginary (thin trace) parts of holomorphic functions z^2 (top left), $\frac{1}{z}$ (top right), e^z (bottom left), $\tan z$ (bottom right)

Solution 5

Let $f = u + iv$ and thus compute

$$\nabla u \cdot \nabla v = \partial_x u \partial_x v + \partial_y u \partial_y v,$$

which is 0 if u and v satisfy the Cauchy-Riemann equations. We conclude that the contour lines intersect orthogonally at each point (the gradient of a function is orthogonal at each to the contour lines of such function). $\star$

Problem 6

Let $z = x + iy$ and write x, y as functions of $z, \bar{z}$. Then deduce the Cauchy-Riemann operators by using the chain rule and treating $z, \bar{z}$ as independent variables. Verify that the Cauchy-Riemann operator $\partial_{\bar{z}}$ “differentiates with respect to $\bar{z}$ only” by showing that

- i) $\partial_{\bar{z}} \bar{z}^n = n \bar{z}^{n-1}$ for $n \in \mathbb{N}$;
- ii) $\partial_{\bar{z}}(\sin \bar{z} \cos z) = \cos \bar{z} \cos z$;
- iii) $\partial_{\bar{z}}(z^2 \bar{z}) = z^2$;
- iv) $\partial_{\bar{z}} e^{z\bar{z}} = z e^{z\bar{z}}$.

Solution 6

i) Straightforward, use the power rule for real differentiation.

ii) Compute $\frac{1}{2}(\partial_x + i\partial_y)(\sin(x - iy) \cos(x + iy))$

$$\begin{aligned}
&= \frac{1}{2}(\cos(x - iy) \cos(x + iy) - \sin(x - iy) \sin(x + iy) + i[-i \cos(x - iy) \cos(x + iy) \\
&\quad - i \sin(x - iy) \sin(x + iy)]) \\
&= \cos(x - iy) \cos(x + iy) = \cos \bar{z} \cos z.
\end{aligned}$$

iii) Straightforward.

iv) Use the properties of the exponential function and differentiate, the result follows directly. $\star$

Problem 7

Show that the function

$$f(z) = \begin{cases} z^5/|z|^4 & z \neq 0, \\ 0 & z = 0, \end{cases}$$

satisfies the Cauchy Riemann equations at $z = 0$, but at the same point it is not differentiable.

Solution 7

By definition, we have that (identify u, v via $f = u + iv$)

$$f(z) = \frac{(x + iy)^5}{(x^2 + y^2)^2}$$

for $z \neq 0$, and $f(0) = 0$ so that

$$\partial_x u(0, 0) = \lim_{h \rightarrow 0} \frac{u(h, 0) - u(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h^5/h^4}{h} = 1$$

and $(f(ih) = iy^5/y^4 \implies v(0, h) = h^5/h^4)$

$$\partial_y v(0, 0) = \lim_{h \rightarrow 0} \frac{v(0, h) - v(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{v(0, h) - v(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h^5/h^4}{h} = 1,$$

and hence $\partial_x u(0, 0) = \partial_y v(0, 0)$ and similarly, $\partial_y u(0, 0) = -\partial_x v(0, 0) = 0$. Nevertheless, taking the definition of complex derivative we see that

$$\lim_{w \rightarrow 0} \frac{f(w) - f(0)}{w} = \lim_{w \rightarrow 0} \frac{w^5}{|w|^4 w} = \lim_{w \rightarrow 0} \frac{w^4}{|w|^4}$$

which for w approaching from the real line yields 1 and approaching from the curve $y = x$ we have terms of the form $x + ix$ and hence the limit becomes

$$\lim_{w \rightarrow 0} \frac{x^4(1+i)^4}{4x^4} = \frac{(1+i)^4}{4} = -1.$$

We therefore conclude that f is not complex differentiable at $z = 0$.

